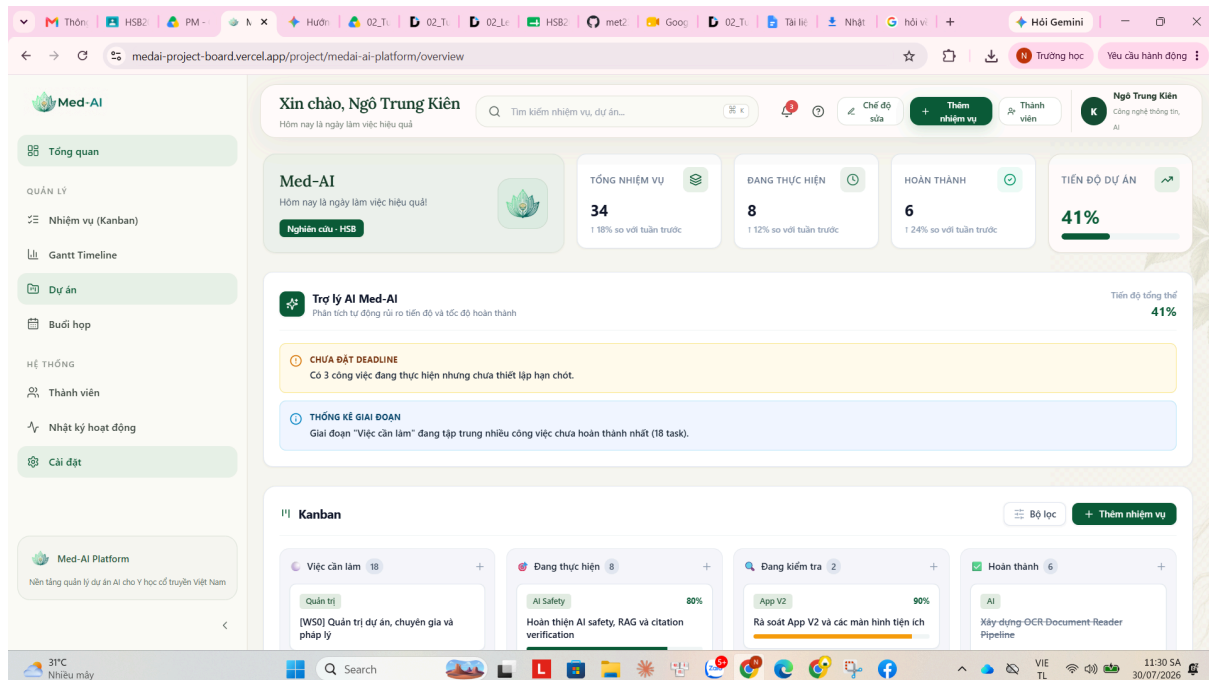


Tutorial 02: PM Tools & Requirements Practice Session

TASK 1: SETUP GITHUB PROJECTS



TASK 2: REQUIREMENTS ANALYSIS (PHÂN TÍCH YÊU CẦU DỰ ÁN TO-DO APP)

1. Yêu cầu chức năng (3 Functional Requirements - FR)

FR1 (Thêm công việc): Hệ thống cho phép người dùng nhập nội dung và thêm một công việc mới vào danh sách.

FR2 (Đánh dấu hoàn thành): Hệ thống cho phép người dùng bấm chọn để chuyển trạng thái công việc sang đã hoàn thành.

FR3 (Xóa công việc): Hệ thống cho phép người dùng xóa một công việc ra khỏi danh sách.

2. Yêu cầu phi chức năng (2 Non-Functional Requirements - NFR)

NFR1 (Giao diện - Usability): Giao diện ứng dụng phải đáp ứng hiển thị mượt mà trên cả thiết bị di động và máy tính (Mobile Responsive).

NFR2 (Hiệu năng - Performance): Thời gian phản hồi của hệ thống khi thực hiện các thao tác (thêm, xóa, sửa) phải dưới 1.5 giây.

User Story 1: Thêm công việc mới

User Story: As a user, I want to add a new task so that I can keep track of my daily work.

Acceptance Criteria: Khi người dùng nhập tên công việc vào ô văn bản và nhấn nút "Add", công việc mới phải hiển thị ngay ở đầu danh sách với trạng thái chưa hoàn thành.

User Story 2: Đánh dấu công việc hoàn thành

User Story: As a user, I want to mark a task as completed so that I know which tasks are finished.

Acceptance Criteria: Khi bấm vào ô checkbox bên cạnh công việc, dòng chữ của công việc đó sẽ bị gạch ngang và chuyển sang màu xám.

User Story 3: Xóa công việc

User Story: As a user, I want to delete a task so that I can remove unwanted entries from my list.

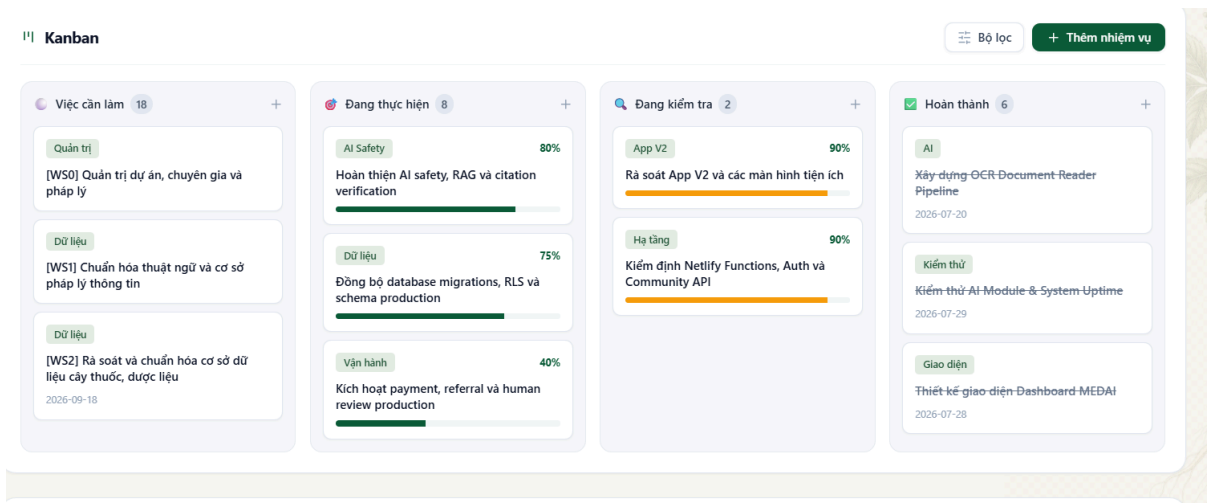
Acceptance Criteria: Khi người dùng nhấn vào nút "Delete" (biểu tượng thùng rác), hệ thống hiển thị thông báo xác nhận và xóa vĩnh viễn công việc khỏi danh sách sau khi người dùng đồng ý.

TASK 4: POPULATE THE PM TOOL (CẬP NHẬT DỮ LIỆU LÊN GITHUB)

Thực hiện đưa các User Stories vào công cụ quản lý dự án GitHub.

Các bước thực hiện

1. Vào mục Issues trên Repository, tạo 3 Issue mới tương ứng với 3 User Stories ở Task 3.
2. Đưa cả 3 Issue này vào bảng GitHub Projects tại cột To Do.
3. Chọn 1 Issue (User Story 1), thực hiện gán người phụ trách (Assignee) cho chính tài khoản GitHub của mình, sau đó kéo thả Issue này sang cột In Progress.



BẢNG KIỂM TRA HOÀN THÀNH (VALIDATION CHECKLIST)

- [x] GitHub Project board is active: Bảng Kanban đã được kích hoạt và hoạt động trên GitHub Projects.
- [x] Board has at least 3 issues: Bảng đã có đủ 3 thẻ nhiệm vụ tương ứng 3 User Stories.
- [x] Issues follow the standard User Story format: Các Issue đều được viết đúng cấu trúc chuẩn As a... I want to... so that...
- [x] At least one issue is in the "In Progress" column: Đã chuyển 1 Issue sang cột In Progress và phân công cho bản thân.

PROJECT MANAGEMENT AND REQUIREMENTS ENGINEERING REPORT

- Course: Practical Software Development Studio (HSB2006)
- Session: Session 02 - PM Tools & Requirements
- Author: Ngo Trung Kien
- Sample Project: Web-based To-Do Application
- Tools Used: GitHub Projects and GitHub Issues

1. THEORETICAL FOUNDATION

Agile methodology provides an iterative approach to software development, enabling teams to deliver incremental value rapidly. Within this paradigm, Scrum organizes work into short cycles called Sprints, while Kanban visualizes the workflow across distinct state columns such as To Do, In Progress, and Done.

Requirements engineering serves as the foundation of software planning. Functional Requirements define specific system capabilities, whereas Non-Functional Requirements establish operational quality standards such as performance, usability, and responsiveness. User Stories articulate these requirements from the perspective of end users using the standard structure "As a [role], I want to [action] so that [benefit]", complemented by Acceptance Criteria that define completion conditions.

2. TASK 1: SETUP GITHUB PROJECTS

A new project board was initialized in the GitHub repository under the Projects tab. The Kanban template was selected and configured with three standard workflow columns: To Do, In Progress, and Done.

[INSERT IMAGE 1: Screenshot of your active GitHub Projects Kanban board showing the three columns To Do, In Progress, and Done]

3. TASK 2: REQUIREMENTS ANALYSIS

The requirements analysis for the web-based To-Do application identified three Functional Requirements and two Non-Functional Requirements.

Functional Requirements

1. Functional Requirement 1: The system must allow users to input text and add a new task to the list.
2. Functional Requirement 2: The system must allow users to toggle a task status to mark it as completed.
3. Functional Requirement 3: The system must allow users to remove an existing task from the list.

Non-Functional Requirements

1. Non-Functional Requirement 1 (Usability): The web application user interface must be mobile responsive across desktop and mobile browsers.
2. Non-Functional Requirement 2 (Performance): The system must process user interactions such as adding, completing, or deleting tasks with a response time under 1.5 seconds.

4. TASK 3: WRITING USER STORIES AND ACCEPTANCE CRITERIA

The identified Functional Requirements were converted into standard User Story formats with associated Acceptance Criteria.

User Story 1: Add New Task

- User Story: As a user, I want to add a new task so that I can keep track of my daily work.
- Acceptance Criteria: When the user types a task name into the input field and clicks the Add button, the new task must immediately appear at the top of the list with an uncompleted status.

User Story 2: Mark Task as Completed

- User Story: As a user, I want to mark a task as completed so that I know which tasks are finished.
- Acceptance Criteria: When the user clicks the checkbox beside a task, the task text must display a line-through style and change to a grayed-out color.

User Story 3: Delete Task

- User Story: As a user, I want to delete a task so that I can remove unwanted entries from my list.
- Acceptance Criteria: When the user clicks the Delete button next to a task, a confirmation prompt appears, and confirming it permanently removes the item from the list.

5. TASK 4: POPULATE THE PM TOOL

Each User Story was created as an Issue in the GitHub repository and added to the GitHub Project board under the To Do column. User Story 1 was then assigned to the project owner and moved to the In Progress column.

[INSERT IMAGE 2: Screenshot of your GitHub Repository Issues tab displaying all three created User Story issues]

[INSERT IMAGE 3: Screenshot of your GitHub Projects Kanban board showing two issues in To Do and one assigned issue in In Progress]

6. VALIDATION CHECKLIST AND BEST PRACTICES

Validation Checklist

1. GitHub Project board is active and fully configured.
2. Board contains three distinct User Story issues.
3. Issues strictly follow the standard User Story format.
4. At least one issue is assigned and located in the In Progress column.

Summary of Best Practices

1. Avoid vague requirements by establishing measurable criteria like response time limits.
2. Ensure Acceptance Criteria are defined before starting task development.
3. Maintain daily updates on the Kanban board to reflect accurate project progress.